

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Uniform indentation maintained throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Files are organized into templates, static, models, and app folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables are named clearly based on their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions have meaningful and descriptive names.
5	Code Comments	Inline and block comments explain logic	Yes	Important sections are documented with comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	Code is divided into reusable modules and functions.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate and unnecessary code has been removed.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling are implemented.

Reusable Components / Modules:

S. No	Component / Module Name	Language / Technology	Y Where Reused	Reusability Level (High / Medium / Low)
1	Loan Prediction Module	Python (Flask, XGBoost)	Prediction & Testing	High

2	Data Preprocessing	Python (Pandas , numPy)	Model Training	High
3	Web Interface	HTML, CSS, Bootstrap	User Dashboard	Medium
4	Model Training Script	Scikit-learn, XGBoost	Training & evaluation	High
5	Input Validation	Python	User Input processing	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well structure and organized
Readability	5	Easy to read with meaningful naming
Reusability	5	Reusable modules and function
Documentation / Comments	4	Sufficient comments added
Overall Score	5/5	Clean, maintainable and reusable code